HW2.3

SCIE 001 Math

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1. Recall that the probability density function for the electron-proton distance r in the hydrogen atom ground state (i.e., the 1s orbital) is given by

$$\begin{aligned} f(r) &= 4\pi r^2 \psi_{100}^2(r) \\ &= Ar^2 e^{\frac{-2r}{a_0}} \end{aligned}$$

where a_0 is the Bohr radius.

(a) Find the cumulative distribution function $F(r)$.

From the definition of the cumulative distribution function $F(r)$:

$$F(r) = \int_{-\infty}^{r} f(x)\,\mathrm{d}x$$

So we can just compute the improper integral. In our case, the electron-proton distance cannot be negative, so the integral spans from 0 to ∞.

$$F(r) = \int_0^r f(x)\,\mathrm{d}x$$

$$F(r) = \int_0^r Ax^2 e^{\frac{-2x}{a_0}}\,\mathrm{d}x$$

We can do integration by parts:

$$\begin{aligned} u &= Ax^2 & \mathrm{d}v &= e^{\frac{-2x}{a_0}}\,\mathrm{d}x \\ \mathrm{d}u &= 2Ax\,\mathrm{d}x & v &= \frac{a_0}{-2} e^{\frac{-2x}{a_0}} \end{aligned}$$

$$F(r) = \left(Ax^2\right)\left(\frac{a_0}{-2} e^{\frac{-2x}{a_0}}\right)\Big]_0^r - \int_0^r \frac{a_0}{-2} e^{\frac{-2x}{a_0}} 2Ax\,\mathrm{d}x$$

$$F(r) = \left(-\frac{1}{2} A a_0 x^2 e^{\frac{-2x}{a_0}}\right)\Big]_0^r - \int_0^r -a_0 Axe^{\frac{-2x}{a_0}}\,\mathrm{d}x$$

Doing integration by parts again:

$$\begin{aligned} t &= -a_0 Ax & \mathrm{d}p &= e^{\frac{-2x}{a_0}}\,\mathrm{d}x \\ \mathrm{d}t &= -a_0 A\,\mathrm{d}x & p &= \left(\frac{a_0}{-2}\right) e^{\frac{-2x}{a_0}} \end{aligned}$$

So we get:

$$F(r) = \left(-\frac{1}{2}Aa_0x^2e^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \left(\left((-a_0Ax)\left(\left(\frac{a_0}{-2}\right)e^{\frac{-2x}{a_0}}\right)\right)\Bigg]_0^r - \int_0^r \left(\frac{a_0}{-2}\right)e^{\frac{-2x}{a_0}}(-a_0A)\,\mathrm{d}x\right)$$

$$F(r) = \left(-\frac{1}{2}Aa_0x^2e^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \left(\left(\frac{1}{2}a_0^2Axe^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \int_0^r \left(\frac{a_0^2A}{2}\right)e^{\frac{-2x}{a_0}}\,\mathrm{d}x\right)$$

$$F(r) = \left(-\frac{1}{2}Aa_0x^2e^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \left(\left(\frac{1}{2}a_0^2Axe^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \left(\left(\frac{a_0^2A}{2}\right)\left(\frac{a_0}{-2}\right)e^{\frac{-2x}{a_0}}\right)\Bigg]_0^r\right)$$

$$F(r) = \left(-\frac{1}{2}Aa_0x^2e^{\frac{-2x}{a_0}}\right)\Bigg]_0^r - \left(\frac{1}{2}a_0^2Axe^{\frac{-2x}{a_0}}\right)\Bigg]_0^r + \left(\frac{1}{-4}a_0^3Ae^{\frac{-2x}{a_0}}\right)\Bigg]_0^r$$

$$\begin{aligned} F(r) = &\left(\left(-\frac{1}{2}Aa_0r^2e^{\frac{-2(r)}{a_0}}\right) - \left(-\frac{1}{2}Aa_00^2e^{\frac{-2(0)}{a_0}}\right)\right) \\ &-\left(\left(\frac{1}{2}a_0^2A(r)e^{\frac{-2(r)}{a_0}}\right) - \left(\frac{1}{2}a_0^2A(0)e^{\frac{-2(0)}{a_0}}\right)\right) \\ &+\left(\left(\frac{1}{-4}a_0^3Ae^{\frac{-2(r)}{a_0}}\right)\right) - \left(\left(\frac{1}{-4}a_0^3Ae^{\frac{-2(0)}{a_0}}\right)\right)\end{aligned}$$

Which simplifies to:

$$F(r) = -\frac{1}{2}Aa_0r^2e^{\frac{-2r}{a_0}} - \frac{1}{2}a_0^2Are^{\frac{-2r}{a_0}} - \frac{1}{4}a_0^3Ae^{\frac{-2r}{a_0}} + \frac{1}{4}a_0^3A$$

(b)

> Determine the value of A (i.e., “normalize” the distribution).

To normalize the function, we can set its indefinite integral equal to 1, and solve for A.

$$\int_0^\infty f(x)\,\mathrm{d}x = 1$$

$$\int_0^\infty Ax^2e^{\frac{-2x}{a_0}}\,\mathrm{d}x = 1$$

Fortunately, we already computed the integral in question 1.a), so we can rewrite this integral in terms of the cumulative distribution function.

$$\int_0^\infty f(x)\,\mathrm{d}x = \lim_{r\to\infty}\int_0^r f(x)\,\mathrm{d}x = \lim_{r\to\infty} F(r) = 1$$

So we can write:

$$\begin{aligned} 1 &= \lim_{r\to\infty} F(r) \\ 1 &= \lim_{r\to\infty}\left(\frac{1}{2}Aa_0r^2e^{\frac{-2r}{a_0}} - \frac{1}{2}a_0^2Are^{\frac{-2r}{a_0}} - \frac{1}{4}a_0^3Ae^{\frac{-2r}{a_0}} + \frac{1}{4}a_0^3A\right) \\ 1 &= \lim_{r\to\infty}\left(\frac{1}{2}Aa_0r^2e^{\frac{-2r}{a_0}}\right) - \lim_{r\to\infty}\left(\frac{1}{2}a_0^2Are^{\frac{-2r}{a_0}}\right) - \lim_{r\to\infty}\left(\frac{1}{4}a_0^3Ae^{\frac{-2r}{a_0}}\right) + \lim_{r\to\infty}\left(\frac{1}{4}a_0^3A\right) \\ 1 &= \lim_{r\to\infty}\left(\frac{Aa_0r^2}{2e^{\frac{2r}{a_0}}}\right) - \lim_{r\to\infty}\left(\frac{Aa_0^2r}{2e^{\frac{2r}{a_0}}}\right) - 0 + \frac{1}{4}a_0^3A\end{aligned}$$

Now we do L’Hôpital’s rule on both the limits:

$$1 = \lim_{r\to\infty}\left(\frac{2Aa_0r}{2(\frac{a_0}{2})e^{\frac{2r}{a_0}}}\right) - \lim_{r\to\infty}\left(\frac{Aa_0^2}{2(\frac{a_0}{2})e^{\frac{2r}{a_0}}}\right) + \frac{1}{4}a_0^3A$$

$$1 = \lim_{r\to\infty}\left(\frac{2Aa_0r}{2(\frac{a_0}{2})e^{\frac{2r}{a_0}}}\right) - 0 + \frac{1}{4}a_0^3A$$

Doing L'Hopital's rule again on the limit:

$$1 = \lim_{r\to\infty}\left(\frac{2Aa_0}{2(\frac{a_0}{2})(\frac{a_0}{2})e^{\frac{2r}{a_0}}}\right) + \frac{1}{4}a_0^3A$$

$$1 = 0 + \frac{1}{4}a_0^3A$$

Now solving for A:

$$\frac{4(1)}{a_0^3} = A$$

Therefore:

$$\boxed{A = \frac{4}{a_0^3}}$$

(c)

> Determine the probability that the electron will be found within $2a_0$ of the proton. Does this seem reasonable?

To determine the probability that the electron will be found within $2a_0$ of the proton, we can evaluate the integral:

$$P(r < 2a_0) = \int_0^{2a_0} f(x)\,\mathrm{d}x$$

of the Probability Distribution Function $f(x)$ describing the distance from the particle.

Noting that this is the same evaluating the Cumulative Probability Function $F(x)$ (as found in 1.a)) at $x = a_0$, we can simplify the problem:

$$P(r < 2a_0) = \int_0^{2a_0} f(x)\,\mathrm{d}x = F(2a_0) = -\frac{1}{2}Aa_0(2a_0)^2e^{\frac{-2(2a_0)}{a_0}} - \frac{1}{2}a_0^2A(2a_0)e^{\frac{-2(2a_0)}{a_0}} - \frac{1}{4}a_0^3Ae^{\frac{-2(2a_0)}{a_0}} + \frac{1}{4}a_0^3A$$

Where $A = \frac{4}{a_0^3}$ as found in 1.b). So we can rewrite:

$$\begin{aligned}
P(r < 2a_0) &= -\frac{1}{2}\left(\frac{4}{a_0^3}\right)a_0(2a_0)^2e^{\frac{-2(2a_0)}{a_0}} - \frac{1}{2}a_0^2\left(\frac{4}{a_0^3}\right)(2a_0)e^{\frac{-2(2a_0)}{a_0}} - \frac{1}{4}a_0^3\left(\frac{4}{a_0^3}\right)e^{\frac{-2(2a_0)}{a_0}} + \frac{1}{4}a_0^3\left(\frac{4}{a_0^3}\right) \\
&= -\frac{1}{2}(4)(2)^2e^{(-2(2))} - \frac{1}{2}(4)(2)e^{(-2(2))} - \frac{1}{4}(4)e^{(-2(2))} + \frac{1}{4}(4) \\
&= -(2)(4)e^{-4} - 4e^{-4} - e^{-4} + 1 \\
&= -13e^{-4} + 1
\end{aligned}$$

Therefore the probability that the electron will be found within $2a_0$ of the proton is:

$$P(r < 2a_0) = -13e^{-4} + 1$$

This resembles;

Figure out what this resembles.

(d) Determine the "most probable" and the "expected" radius for the hydrogen atom in the ground state.

This is characterized by the mean of the distribution:

Using the definition of the mean for continuous probability distributions:

$$\mu = \int_{-\infty}^{\infty} x(f(x))\,\mathrm{d}x$$

Since our distribution is 0 at $x < 0$, we can rephrase:

$$\mu = \int_{0}^{\infty} x(f(x))\,\mathrm{d}x$$

Computing the integral:

$$\begin{aligned}\mu &= \int_{0}^{\infty} x(f(x))\,\mathrm{d}x \\ &= \int_{0}^{\infty} xAx^2e^{\frac{-2x}{a_0}}\,\mathrm{d}x \\ &= \int_{0}^{\infty} Ax^3e^{\frac{-2x}{a_0}}\,\mathrm{d}x \\ &= A\int_{0}^{\infty} x^3e^{\frac{-2x}{a_0}}\,\mathrm{d}x\end{aligned}$$

For an integral with such a large exponent on x, it is helpful to use the DI method when doing integration by parts.

Sign	**D**	**I**
$+$	x^3	$e^{\frac{-2x}{a_0}}$
$-$	$3x^2$	$(-\frac{a_0}{2})e^{\frac{-2x}{a_0}}$
$+$	$6x$	$(-\frac{a_0}{2})^2e^{\frac{-2x}{a_0}}$
$-$	6	$(-\frac{a_0}{2})^3e^{\frac{-2x}{a_0}}$
$+$	0	$(-\frac{a_0}{2})^4e^{\frac{-2x}{a_0}}$

Then writing it all out, we get:

$$\mu = A\left((x^3)\left(\left(-\frac{a_0}{2}\right)e^{\frac{-2x}{a_0}}\right) - (3x^2)\left(\left(-\frac{a_0}{2}\right)^2e^{\frac{-2x}{a_0}}\right) + (6x)\left(\left(-\frac{a_0}{2}\right)^3e^{\frac{-2x}{a_0}}\right) - (6)\left(-\frac{a_0}{2}\right)^4e^{\frac{-2x}{a_0}}\right)\Bigg]_0^{\infty}$$

At the limit to infinity, all of these go to zero, as they turn into a L'Hopital's case, resulting in a polynomial on the top that differentiates to 0, and an exponential on the bottom that increase as x goes to infinity.

So we only need to consider the $x = 0$ case:

$$\mu = A\left(-\left(((0)^3)\left(\left(-\frac{a_0}{2}\right)e^{\frac{-2(0)}{a_0}}\right) - (3(0)^2)\left(\left(-\frac{a_0}{2}\right)^2 e^{\frac{-2(0)}{a_0}}\right) + (6(0))\left(\left(-\frac{a_0}{2}\right)^3 e^{\frac{-2(0)}{a_0}}\right) - (6)\left(-\frac{a_0}{2}\right)^4 e^{\frac{-2(0)}{a_0}}\right)\right)$$

Write this better

Using $A = \frac{4}{a_0^3}$ found in part b, we can simplify:

$$\mu = \left(\frac{4}{a_0^3}\right)\left(-\left(((0)^3)\left(\left(-\frac{a_0}{2}\right)e^{\frac{-2(0)}{a_0}}\right) - (3(0)^2)\left(\left(-\frac{a_0}{2}\right)^2 e^{\frac{-2(0)}{a_0}}\right) + (6(0))\left(\left(-\frac{a_0}{2}\right)^3 e^{\frac{-2(0)}{a_0}}\right) - (6)\left(-\frac{a_0}{2}\right)^4 e^{\frac{-2(0)}{a_0}}\right)\right)$$

$$\mu = \left(\frac{4}{a_0^3}\right)\left(-\left(-(6)\left(-\frac{a_0}{2}\right)^4 e^0\right)\right)$$

$$\mu = \left(\frac{4}{a_0^3}\right)(6)\left(\frac{a_0^4}{2^4}\right)$$

$$\mu = (4)(6)\left(\frac{a_0}{2^4}\right)$$

$$\mu = \left(\frac{24}{16}\right)(a_0)$$

Therefore the expected radius is:

$$\mu = \frac{24}{16}a_0$$

Find the mode

2. *Newton's universal law of gravitation* states that the force of attraction between two point masses m and M has magnitude

$$F = \frac{GmM}{r^2}$$

where r is the distance between the masses and $G = 6.67 \times 10^{-11}\ \mathrm{N} \cdot \frac{\mathrm{m}^2}{\mathrm{kg}^2}$ is a constant.

(a) If M represents the mass of the centre of the earth and we regard it as a point mass concentrated at its centre, show that Newton's universal law of gravitation at the earth's surface reduces to $F = mg$, where $g = 9.82\frac{m}{s^2}$. Assume for the calculation of M that the earth is a sphere with radius $6370\ \mathrm{km}$ and mean density $5.52 \times 10^3\, \frac{\mathrm{kg}}{\mathrm{m}^3}$.

First some variable definitions and unit conversion.
Let:

$$R = 6370\ \mathrm{km} = (6370\ \mathrm{km})\left(\frac{1000\ \mathrm{m}}{1\ \mathrm{km}}\right) = 6{,}370{,}000\ \mathrm{m}$$

$$\rho = 5.52 \times 10^3 \frac{\mathrm{kg}}{\mathrm{m}^3}$$

Where R is the radius of the earth, and ρ is the density of the earth.

Approximating the earth as a sphere, its mass is approximately:

$$M = \frac{4}{3}\pi R^3 \rho$$

Approximating the mass of the earth as acting at a point at its center, the distance between any body sufficiently close to the earth's surface would be approximately equal to the radius of the earth. Therefore, from *Newton's universal law of gravitation* the force experienced by such an object would be:

$$F = \frac{GmM}{r^2}$$
$$F = m\frac{GM}{R^2}$$

Calculating $\frac{GM}{R^2}$

$$\begin{aligned}\frac{GM}{R^2} &= \frac{G\left(\frac{4}{3}\pi R^3 \rho\right)}{R^2} \\ &= \frac{4}{3}G\pi R\rho \\ &= \frac{4}{3}(6.67 \times 10^{-11} N \cdot \mathrm{m}^2\mathrm{kg}^{-2})(3.1415926536)(6,370,000\ \mathrm{m})\left(5.52 \times 10^3 \frac{\mathrm{kg}}{\mathrm{m}^3}\right) \\ \frac{GM}{R^2} &= 9.8241040437\frac{\mathrm{m}}{\mathrm{s}^2} \approx 9.82\frac{\mathrm{m}}{\mathrm{s}^2}\end{aligned}$$

$$F = m\frac{GM}{R^2}$$
$$F = m\left(9.82\frac{\mathrm{m}}{\mathrm{s}^2}\right)$$

Therefore, for objects close to earth's surface, *Newton's universal law of gravitation* can be reduced to :

$$F = mg$$

where $g = \left(9.82\frac{\mathrm{m}}{\mathrm{s}^2}\right)$.

(b)
> Use the original $F = \frac{GmM}{r^2}$ with the earth regarded as a point mass to calculate the work required to lift a mass of 10kg from the earth's surface to a height of 10 km.

Let ΔR represent the change in R from its initial height above the earth to its end position.

$$\Delta R = 10\ \mathrm{km} = (10\ \mathrm{km})\left(\frac{1000\ \mathrm{m}}{1\ \mathrm{km}}\right) = 10000\ \mathrm{m}$$

Work W is defined as:

$$W = \int F\,\mathrm{d}r$$

Where F is the force, and $\mathrm{d}r$ is the change in radius.

Therefore we can express this work needed as an integral:

$$W = \int_R^{R+\Delta R} F\,\mathrm{d}r$$

Solving we get:

$$\begin{aligned}
W &= \int_R^{R+\Delta R} \frac{GmM}{r^2}\,\mathrm{d}r \\
&= GmM \int_R^{R+\Delta R} r^{-2}\,\mathrm{d}r \\
&= GmM\left(-r^{-1}\Big]_R^{R+\Delta R}\right) \\
&= GmM(-(R+\Delta R)^{-1} - (-R^{-1})) \\
W &= GmM(-(R+\Delta R)^{-1} + R^{-1})
\end{aligned}$$

As we previously found:

$$M = \frac{4}{3}\pi R^3 \rho$$

$$M = \frac{4}{3}(3.1415926535)(6,370,000\ \mathrm{m})^3\left(5.52\times 10^3 \frac{\mathrm{kg}}{\mathrm{m}^3}\right)$$

Plugging in our values:

$$\begin{aligned}
W &= GmM(-(R+\Delta R)^{-1} + R^{-1}) \\
&= (6.67\times 10^{-11} N\cdot \mathrm{m}^2\mathrm{kg}^{-2})(10\ \mathrm{kg})\times \\
&\left(\frac{4}{3}(3.1415926535)(6,370,000\ \mathrm{m})^3\left(5.52\times 10^3\frac{\mathrm{kg}}{\mathrm{m}^3}\right)\right)\times \\
&(-((6,370,000\ \mathrm{m}) + 10000\ \mathrm{m})^{-1} + (6,370,000\ \mathrm{m})^{-1}) \\
W &= 980870.576\ \mathrm{J}
\end{aligned}$$

(c)

> Calculate the work in part (b) using the constant gravitational force $F = mg$ in part (a). Is there a significant difference?

As in part (b), let ΔR represent the change in R from its initial height above the earth to its end position.

$$\begin{aligned}
W &= \int_R^{R+\Delta R} F\,\mathrm{d}r \\
&= \int_R^{R+\Delta R} mg\,\mathrm{d}r \\
&= mgr\Big]_R^{R+\Delta R} \\
&= mg(R+\Delta R) - mg(R) \\
W &= mg\Delta R
\end{aligned}$$

Using the value of g we calculated from part a:

$$g = 9.8241040437\frac{\mathrm{m}}{\mathrm{s}^2}$$

$$\Delta R = 10000\ \mathrm{m}$$

$$\begin{aligned} W &= mg\Delta R \\ &= (10\ \mathrm{kg})\left(9.8241040437\frac{\mathrm{m}}{\mathrm{s}^2}\right)(10000\ \mathrm{m}) \\ W &= 982410.404\ \mathrm{J} \end{aligned}$$

Calculating the difference:

$$|982410.404\ \mathrm{J} - 980870.576\ \mathrm{J}| = 1539.828\ \mathrm{J}$$

Dividing by the integrated form of work, and multiplying by 100%, we get a relative difference of:

$$\frac{1539.828\ \mathrm{J}}{980870.576\ \mathrm{J}} \times 100\% = 0.157\%$$

Therefore there is a 0.157% difference between the values, showing an insignificant difference.

3. In the next problem you will be working with a couple of examples of a fractal. A fractal is a mathematical set that displays a self-similarity property; that is, it exhibits a repeating pattern that displays at every scale.

(a) The Cantor set, named after the German mathematician Georg Cantor (1845-1918), is constructed as follows. We start with the closed interval $[0, 1]$ and remove the open interval $\left(\frac{1}{3}, \frac{2}{3}\right)$. That leaves the two intervals $\left[0, \frac{1}{3}\right]$ and $\left[\frac{2}{3}, 1\right]$. We then remove the open middle interval third of each of these two remaining intervals. We continue this procedure indefinitely, at each step removing the open middle third of every interval that remains from the preceding step. The Cantor set consists of the numbers that remain in the original interval $[0, 1]$ after all those intervals have been removed.

i. Sketch a diagram that shows the first 5 iterations of the Cantor set.

Where each bar is a closed sub-interval of $[0, 1]$.

I wrote this myself using a recursive algorithm in the typst programming language. The code is below:

```
#let draw_cantor(x) = {
  if x == 0 { // base case for the recursive process
    line(length: 100%, stroke: 10pt,) // just draw the line.
  } else {
    let h = draw_cantor(x - 1);
    grid(columns:(1fr, 1fr, 1fr),h,[],h,)
  }
}
#for value in range(0, 10) {
```

```
  draw_cantor(value) // call the draw_cantor function 5 times, each with a
}
```

Include a handwritten drawing as well

ii. Give examples (at least 5) of some numbers in the Cantor set.

Since we are always removing the open interval in the middle of a closed interval, the numbers defining closed bounds themselves are never removed. Therefore we can just take the first 5 bounds.

At each level of the set, to get the next level, we remove the middle third of each interval. Therefore to get the next set of bounds, we can add one third the width of each interval to each lower bound, and subtract one third from each upper bound.

Bounds	One-third the Width of each Interval
$[0,1]$	$\frac{1}{3}$
$\left[0,\frac{1}{3}\right]\cup\left[\frac{2}{3},1\right]$	$\frac{1}{9}$
$\left[0,\frac{1}{9}\right]\cup\left[\frac{2}{9},\frac{1}{3}\right]\cup\left[\frac{2}{3},\frac{7}{9}\right]\cup\left[\frac{8}{9},1\right]$	$\frac{1}{27}$

This leaves us with our final list:

$$0,\frac{1}{9},\frac{2}{9},\frac{1}{3},\frac{2}{3},\frac{7}{9},\frac{8}{9},1,$$

are all in the cantor set.

iii. Show that the total length of all the intervals that are removed is 1. Despite that, the Cantor set is not an empty set.

In the first level of recursion, we remove the open middle third of the interval, spanning a length of $\frac{1}{3}$, for a total of $\frac{1}{3}$ length removed.

In the second level of recursion we remove the open middle third of the two sub-intervals we just created, each spanning a length of $\frac{1}{9}$, for a total of $\frac{2}{9}$ length removed.

For the n^{th} level of recursion, we remove the middle third of 2^{n-1} intervals, spanning a length of $\frac{1}{3^n}$. Therefore, to get the total length removed, we can simply take the sum of these removals for an infinite recursion depth.

Clean up the math by simplifying the reasoning like the sierpinski carpet one.

$$L=\sum_{n=1}^{\infty}2^{n-1}\left(\frac{1}{3^n}\right)=\sum_{n=1}^{\infty}\frac{2^{n-1}}{3^n}$$

Where L represents the length removed.

Now we can show that this infinite series converges to a value.

With some algebraic manipulation, we can form this into a geometric series:

$$\begin{aligned} L &= \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} \\ &= \sum_{n=1}^{\infty} \frac{2^{n-1}}{3(3^{n-1})} \\ &= \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^{n-1} \end{aligned}$$

Since $\left|\frac{2}{3}\right| < 1$, we can use the formula:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

So:

$$L = \sum_{n=1}^{\infty} \left(\frac{1}{3}\right)\left(\frac{2}{3}\right)^{n-1} = \frac{\frac{1}{3}}{1-\left(\frac{2}{3}\right)} = \frac{\left(\frac{1}{3}\right)}{\left(\frac{1}{3}\right)} = 1$$

Hence, the length of removed is equal to 1.

(b)
> The Sierpinski carpet is a two-dimensional counterpart of the Cantor set. It is constructed by removing the centre one-ninth of a square of side 1, then removing the centres of the eight smaller remaining squares, and so on. A visualization of the Sierpinski carpet is available on Wikipedia (https://en.wikipedia.org/wiki/Sierpinski_carpet).

i.
> Show that the sum of the areas of the removed squares is 1. This implies that the Sierpinski carpet has area 0.

Just like problem a)ii., we can construct a series representing the area removed.

In the first level of recursion we remove the center one-ninth of one square of area 1, for a total of $\left(\frac{1}{9}\right)(1)(1)$ area removed.

In the second level of recursion we remove the center one-ninth of 8 squares of area $\frac{1}{9}$, for a total of $\left(\frac{1}{9}\right)(8)\left(\frac{1}{9}\right)$ area removed.

For the n^{th} level of recursion, we remove the center one-ninth of 8^{n-1} squares of area $\frac{1}{9^{n-1}}$, for a total of $\left(\frac{1}{9}\right)(8^{n-1})\left(\frac{1}{9^{n-1}}\right)$

Therefore, to get the total area removed, we can simply take the sum of these removals for an infinite recursion depth.

$$A = \sum_{n=1}^{\infty} (8^{n-1})\left(\frac{1}{9^{n-1}}\right)$$

Where A represents the length removed.

Now we can show that this infinite series converges.

Similar to the previous question, we can rearrange it into a geometric series:

$$A = \sum_{n=1}^{\infty} \frac{8^{n-1}}{9^{n-1}}$$
$$= \sum_{n=1}^{\infty} \left(\frac{8}{9}\right)^{n-1}$$

Since $\left|\frac{8}{9}\right| < 1$, we can use the formula:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}$$

So:

$$A = \sum_{n=1}^{\infty} \left(\frac{1}{9}\right) \left(\frac{8}{9}\right)^{n-1} = \frac{\frac{1}{9}}{1 - \left(\frac{8}{9}\right)} = \frac{\left(\frac{1}{9}\right)}{\left(\frac{1}{9}\right)} = 1$$

Therefore the total area removed is 1.